ASSIGNMENT-1 SOLUTION

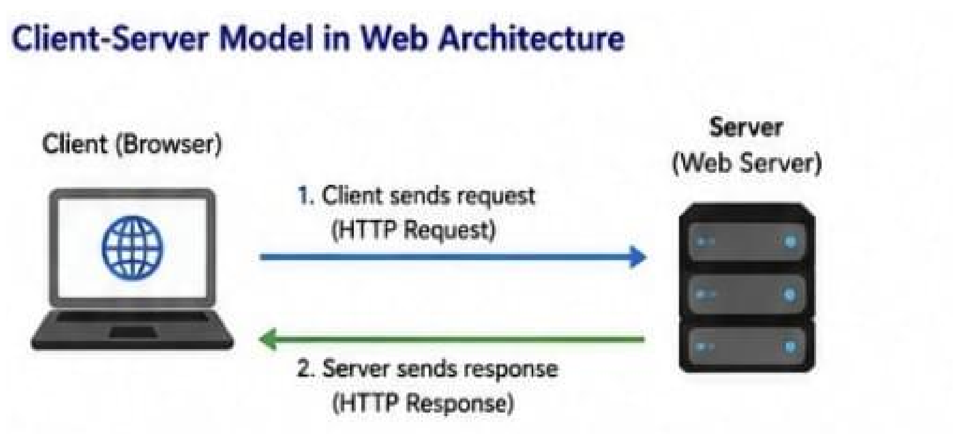
Q1. Explain the difference between frontend, backend, and full-stack development with
suitable real-world examples.

ANS :-

• Frontend Development: Deals with the part of the website
that users see and interact with. It includes designing layout, UI elements, and ensuring a good user experience.
Technologies: HTML, CSS, JavaScript, React, Angular, Vue.js Example: The login page of an e-commerce site where users enter their email and password.
• Backend Development: Deals with the server side, database,
and application logic. It handles data processing, authentication,
and business logic.
Technologies: Node.js, Python, PHP, Java, Express, Django
Example: Processing the login request, checking credentials in database, and returning a response.
• Full-Stack Development: Combination of both frontend and
backend development. A full-stack developer can build complete web applications.
Technologies: HTML, CSS, JavaScript, React, Node.js, MongoDB Example: Building an e-commerce website including UI, server, API, and database.

Q2. Create a simple diagram showing how the client-server model works in web architecture.

ANS:-



Q3. Describe how a browser requests and displays a web page from a web server.

ANS:-

1. User enters URL (e.g., https://www.example.com) in the browser.
2. Browser sends an HTTP request to the web server for that URL.
3. Web server processes the request and sends back an HTP response containing HTML, CSS, JavaScript, images, etc
4. Browser parses the HTML to create the DOM (Document Object Model)
5. Browser parses the CSS to create the CSSOM (CSS Object Model).
6. Browser combines DOM and CSSOM to render the page.
7. JavaScript is executed to make the page interactive.
8. Finally, the page is displayed to the user.

Q4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

ANS:-

Tools Required for Web Development Environment

Tool	Purpose
Visual Studio Code (VS Code)	Code editor used to write and manage HTML, CSS, JavaScript and other languages.
Google Chrome	Web browser used for running and testing web pages.
Live Server (Extension)	VS Code extension that provides live reloading of web pages during development.
Prettier (Extension)	Code formatter that helps maintain consistent code style.
Git & GitHub	Version control and code hosting platform.
Node.js & npm	For running JavaScript on server side and managing packages.

Q5. Explain what a web server is and give examples of commonly used servers.

ANS:- A web server is a software (or hardware) that stores, processes anc delivers web pages to clients (browsers) over the internet using the HTTP protocol.
Examples of commonly used web servers:
• Apache HTTP Server
NGIMX
Nginx
Microsoft IS (Internet Information Services)
LiteSpeed
Caddy Server

Q6. Define the roles of a frontend developer, backend developer, and database administrator
in a project.

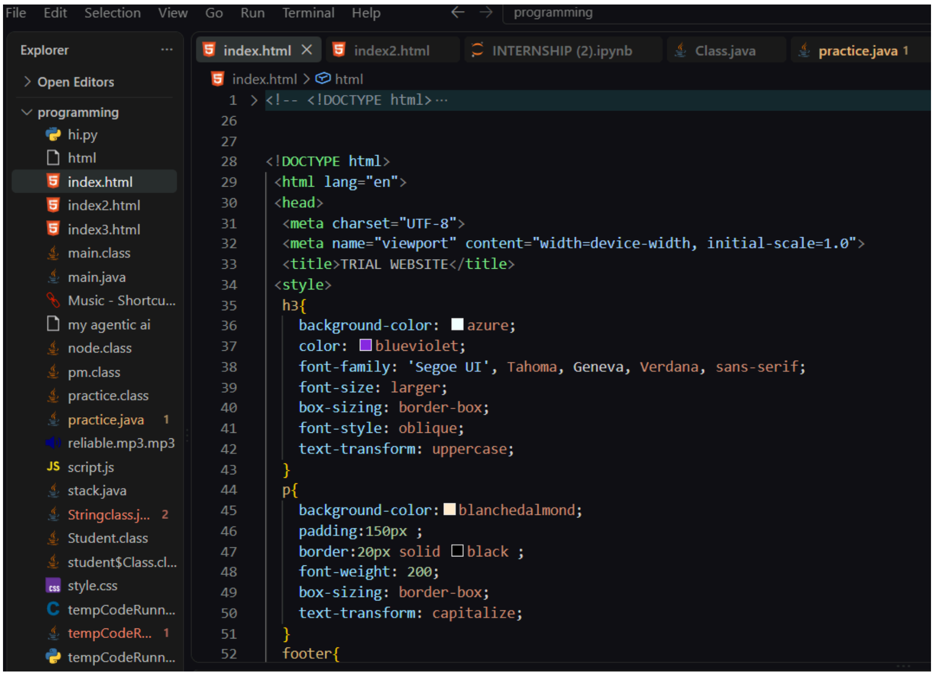
ANS:-

. Roles in a Web Development Project

Role	Responsibilities
Frontend Developer	Builds the user interface, ensures responsive design, and works on user experience. Works with HTML, CSS, JavaScript, frameworks like React, Angular.
Backend Developer	Builds the server, APIs, handles business logic, authentication, and integrates database. Works with Node.js, Python, PHP, Java, etc.
Database Administrator (DBA)	Designs database schema, manages data storage, ensures data security, performance and backup/recovery. Works with MySQL, PostgreSQL, MongoDB, etc.

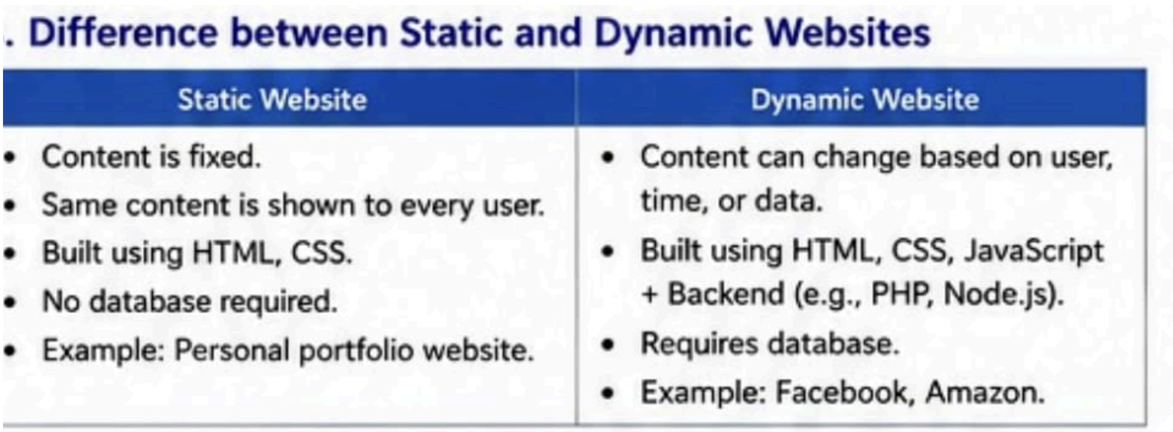
Q7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

ANS:-



Q8. Explain the difference between static and dynamic websites. Provide an example of each.

ANS:-



. **Difference between Static and Dynamic Websites**

Static Website	Dynamic Website
• Content is fixed. • Same content is shown to every user. • Built using HTML, CSS. • No database required. • Example: Personal portfolio website.	• Content can change based on user, time, or data. • Built using HTML, CSS, JavaScript + Backend (e.g., PHP, Node.js). • Requires database. • Example: Facebook, Amazon.

Q9. Research and list five web browsers. Explain how rendering engines differ between
them.

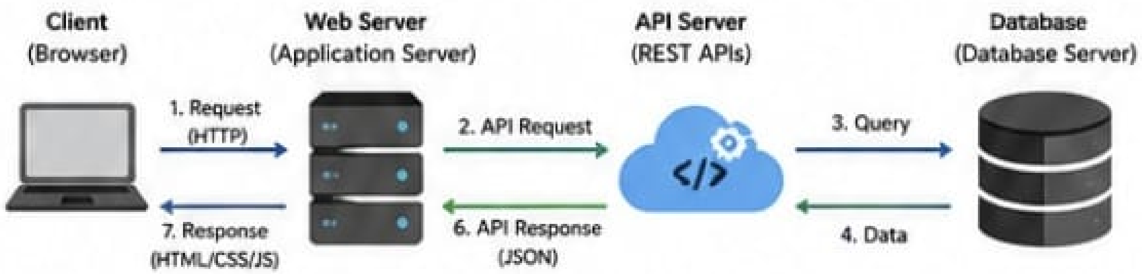
ANS:-

Browser	Rendering Engine	Description
Google Chrome	Blink	Fast rendering engine, used in Chrome and Edge.
Mozilla Firefox	Gecko	Open-source engine used in Firefox.
Microsoft Edge	Blink	New Edge is based on Chromium (Blink engine).
Apple Safari	WebKit	Engine used in Safari, optimized for Apple devices.
Opera	Blink	Opera also uses Blink engine for rendering.

Q10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

ANS:-

10. Basic Web Architecture Flow (Client, Server, Database, APIs)



1. Client requests a page or data.
2. Web server calls the API.
3. API server queries the database.
4. Database returns the data.
5. API sends data back to web server.
6. Web server processes and sends response to client.
7. Browser renders and displays the page.